



A Study on the Effectiveness of Current Data in Motor Mechanical Fault Diagnosis Using XAI

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Received: 6 March 2022 / Revised: 6 July 2022 / Accepted: 22 July 2022 / Published online: 18 August 2022
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Abstract

Vibration and noise data are signal that directly affected by the motor's mechanical fault. However, the stator current is also affected to some extent. A deep learning can classify little changes. In other words, the current is classifiable. When using XAI (eXplainable A.I) which outputs the data effect of the trained model, it is available to know the feature points of the current from the model's view. In this study, the classical ideal feature point and the XAI-LRP output are compared. And when a XAI is inapplicable (e.g. model cannot be trained), an Pearson correlation coefficient analysis method is proposed. Through this study, the effectiveness of current data in mechanical fault diagnosis is confirmed.

Keywords Motor fault diagnosis · Current data analysis · XAI · Machine learning

1 Introduction

The noise and vibration data are mainly used for diagnosing motor faults. Stator current data is commonly used for electrical fault diagnosis such as short-circuit fault. Fast responsiveness is required in electrical faults. Inverter fault diagnosis is field using current data. However, the signals that directly indicate mechanical faults, gear wear or bearing broken, are noise and vibration. The vibration data is more robust than the noise. Therefore, it is preferred in the diagnosis. The current is indirectly affected by mechanical fault. The load changes influences the current signal. When the mechanical fault is occurred not enough to affect the load change, fault diagnosis using current signal is not suitable. However, the research on fault diagnosis using current signals has been continuously carried out [1–3]. Motor fault diagnosis with current sensor is carried out in applications with current measuring functions, such as wind turbines, automobiles and helicopters [4]. In the classic studies, it is confirmed the signal changes of current data occurring because gear and bearing fault are affected by

the accessory's characteristic or the motor rotational speed. So, it is possible that mechanical fault can be diagnosed using current data. When diagnose using vibration data, an expensive vibration measurement equipment is required. In addition, if ICP (Integrated Circuit Piezoelectric) type sensor is used, a additional supply circuit and amplification circuit are required. But, if current data is used, the system construction cost may be reduced. The motor control system usually includes a current sensor. So, the cost of the current based diagnosis system is extremely lower than the vibration signal-based system. Also, the current data is more robust rather than the vibration signal. For this reason, study on current signal based motor mechanical fault diagnosis is required.

The deep learning based fault diagnosis is used in this paper. Previous studies have preprocessed the signal features following the classical theory [5]. Preprocessing with classical theory has an impact on improving model performance. The reason for using preprocessing is the deep learning algorithm has black box structure. However, model's classification reason is can be known using XAI algorithm. It is more effective to preprocess the model based on the feature found. Therefore, an experiment is carried out to figure out whether it is possible to diagnose a motor mechanical fault using current data through a deep learning algorithm. Unlike previous studies, the diagnosis model is analyzed through XAI. It is used to evaluate the reliability of the preprocessing through the classical method. In addition, the Pearson correlation coefficient analysis method

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is proposed for when XAI cannot be used. Through this, limitations and availability of the current based mechanical fault diagnosis is analyzed. In addition, data analysis from the model point of view is carried out through LRP analysis of the current signal in the critical fault state.

2 Classical Current Signal Theory

Load changes due to motor mechanical failure are mainly used in the analysis of current signals used in classical fault diagnosis. This is expressed as a variation in the mechanical resistance component. A study is conducted on whether the characteristics of the current signal used for classical fault diagnosis can be used in a method using deep learning. In other words, an experiment is performed to check whether the deep learning algorithm can recognize signal changes caused by load fluctuations. For this, classical fault signal analysis is explained. The main factor that has the biggest influence on the current is definitely the mechanical load. When a gear is broken or a bearing is damaged, a load change of a certain frequency component occurs depending on the rotation frequency of the motor and the level of the fault [6]. This analysis does not enough consider the motor complex fault state. However, it is possible to infer the fault frequency in the complex fault by assuming a linear system. The load change frequency affected by the two components is expressed as (1) [6].

$$f(t) = f_0(t) \pm \sum_{n=1}^N \alpha_n \cdot f_n(t) \pm \sum_{p=1}^P \beta_p \cdot f_p(t) \quad (1)$$

where N, P is the number of components included in one failure element. f_0 is the power source frequency. α_n and β_p are parameters indicating the level of fault. If the fault gets worse, the α_n, β_p values are increased. It is a problem of the classical method that each coefficient according to the failure condition is relatively small. If this value expressed in frequency domain is expressed as load change, it is the same as (2), where f_n and f_p are fault frequencies [7].

$$T_L(t) = T_0(t) \pm \alpha_n \sum_{n=1}^N \sum_{i=1}^{\infty} \cos(2\pi f_n i) \pm \beta_p \sum_{p=1}^P \sum_{i=1}^{\infty} \cos(2\pi f_p i) \quad (2)$$

The load change can be expressed as (2) because the change in the mechanical resistance component due to mechanical fault appears in the form of an impulse train [6]. This is the result of decomposing the impulse train function into a trigonometric function by expressing it as a Fourier series. The frequencies detected in previous method are f_n and f_p . It is determined by the mechanical characteristics of the motor and its rotational speed.

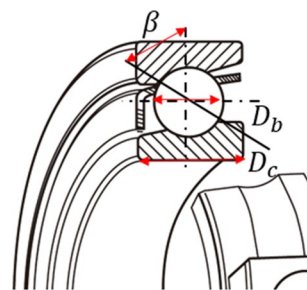


Fig. 1 The structure of ball bearing

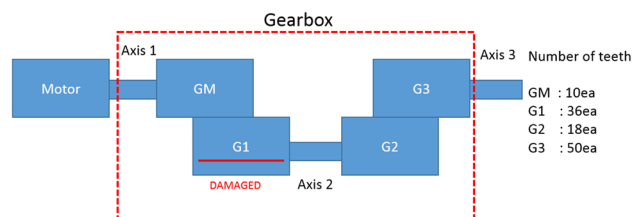


Fig. 2 The gearbox specification

Bearing and gear failures have a significant impact on load changes. Equations (3) and (4) show the fault frequencies of bearing and gear fault [6, 7].

$$GMF = \frac{f_{rm}}{60} \cdot Z \quad (3)$$

$$f_b \begin{cases} f_{outer} = \frac{N_b}{2} f_{rm} \left(1 - \frac{D_b}{D_c} \cos \beta \right) \\ f_{inner} = \frac{N_b}{2} f_{rm} \left(1 + \frac{D_b}{D_c} \cos \beta \right) \\ f_{ball} = \frac{D_c}{D_b} f_{rm} \left(1 - \left(\frac{D_b}{D_c} \cos \beta \right)^2 \right) \end{cases} \quad (4)$$

where f_{rm} [Hz] is the rotation frequency of the motor. D_b and D_c represent the ball diameter and cage diameter, respectively, β represents the contact angle of the ball, and Z is the number of input shaft gear teeth. N_b is the number of balls in the ball bearing. These parameters are shown in Figs. 1 and 2.

Typically, developers know the mechanical specifications of the motor accessory parts. In general, the signal amplitude of the frequency band calculated through (3) and (4) was used for classical fault diagnosis. Predictive maintenance is a technology that determines that a motor abnormality has occurred when the signal level of the calculated frequency band increases by more than a set level. The calculated values are determined by diagnosis experts. However, when deep learning is used for fault diagnosis, there is no need to calculate the failure frequency through (3) or (4) in advance.

The inside of a deep learning model has a black box structure. In general, the performance of a model is evaluated through the classification accuracy of the validation data. In fault diagnosis using deep learning, data preprocessing performance can be improved by knowing the model's judgment criteria. This is to preprocess the data from the model point of view. In addition, the ability to characterize mechanical failure compared to vibration signals can serve as a criterion for determining the limits and validity of weak current signals. If the classically analyzed failure band is the same as the judgment criterion of the deep learning model, the current signal is a signal suitable for failure diagnosis using deep learning. That is, preprocessing according to classical theory is useful. However, if the two bands are different, a better model can be developed by analyzing the model's reference band. In this paper, the characteristics of the current data are analyzed by analyzing the feature points through the XAI technique.

3 Motor Fault Diagnosis Using XAI

The XAI algorithm is a technology that analyzes the inside of a deep learning model with a black box structure. In the previous studies, deep learning model's output is used to evaluate the validity. This means that expert research is needed in the preprocessing process to improve model performance. However, using the XAI algorithm, these studies can be omitted. XAI is a way to ensure the reliability of data from a model point of view. XAI includes a data sensitivity analysis technique that analyzes the impact of a model on data and a technique that infers the validity of the data. The beauty of XAI is that you don't need to build any model to implement it. XAI has a structure that calculates using the values of the trained model. This makes it easy to implement by applying XAI to the trained model.

The most popular method for data sensitivity analysis is SHapley Additive exPlanations (SHAP) proposed by Lloyd Shapley [8]. How much each feature contributed to the model output to generate the overall performance can be expressed numerically using Shapley values. The Shapley value is expressed as (5).

$$\phi_i(v) = \frac{1}{n!} \sum_R [v(P_i^R \cup \{i\}) - v(P_i^R)] \quad (5)$$

where v is the contribution, n is the number of features. P_i^R means all combinations of features other than the i -th data. It can be used in fields such as calculating factors that determine housing prices. However, it is not rational to apply SHAP in the field of fault diagnosis where FFT signals are used rather than categorical data. Since the number of features to be considered increases significantly with the length

of the FFT data, many calculations are required to consider the complex effects. In this process, there is a possibility that the reliability will decrease. It is easier to analyze the frequency bands that directly affect the model when backtracking a model rather than analyzing the sensitivity of the data. The most widely used algorithm among these techniques is LRP (Layer-wise Relevance Propagation) [9]. LRP involves both decomposition processes and relevance propagation. Relevance propagation is the process of decomposing the causes of a particular outcome. The cause of the result is distributed to each weight. Decomposition reduces and dissects the causes obtained by relevance propagation into weights. Finally, the result that can be obtained through LRP is the output of the model that is changed by the change of the input data. Information that can be known when applied to fault diagnosis can know the effect of each frequency of the input signal on the model output. The LRP propagation process is shown in Fig. 3 [10].

The structure of backpropagation used in neural network (NN) training can be used to implement the relevance propagation process. The reference of output in the model is the value of the neuron. Backpropagation propagates the derivative of the previous layer. the relevance propagation Relevance propagation propagates computed relevance. When the contribution propagates along the neuron to the input layer, we know the input neuron's effect on the output. The relevance propagation is same as (6) [8].

$$R_i = \sum_j \frac{a_i w_{ik}}{\sum_h a_h w_{hk}} R_j \quad (6)$$

where j is the number of neurons in the previous layer, h is the number of neurons in the propagating layer, a is the activation function output, i stands for each neuron in the propagating layer. Equation (6) has the form of dividing one multiplication result by the sum of the products of the activation neuron output and the w matrix. This is how the weight is calculated. The LRP algorithm can use the model output as it is. Model weights and activation function outputs such as Relu are values used in model learning and test. Since the operation of (6) is a value that can be immediately known

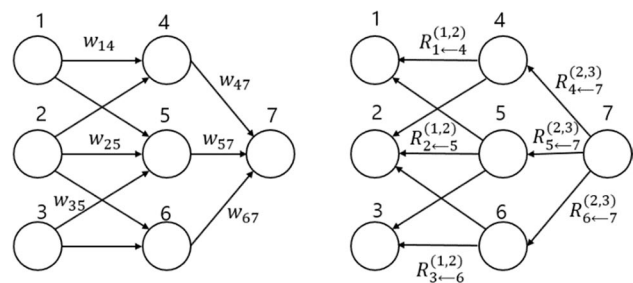


Fig. 3 Propagation flow in LRP

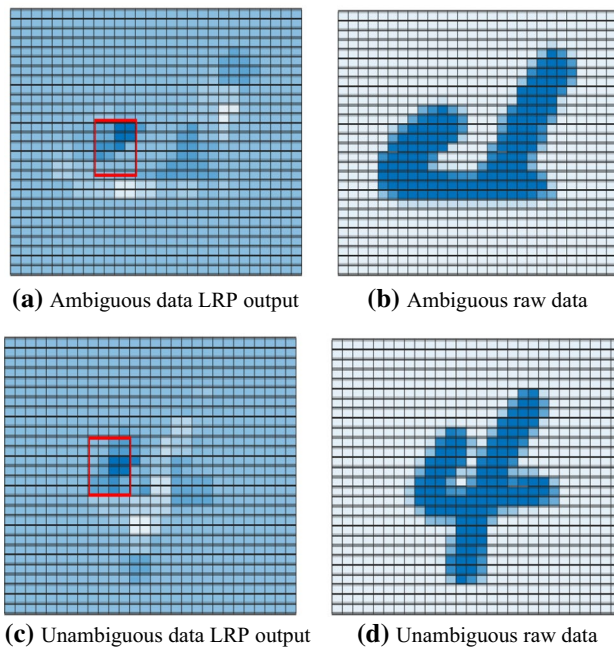


Fig. 4 LRP output of MNIST 4 data. **a, b** Effect of input data and raw image in unclear MNIST 4, respectively. **c, d** Effect of input data and raw image in clear data. The extracted points in **a, c** of this figure are shown in red boxes. By using LRP, classification reference such as the red boxes in **a, c** can be obtained. An experiment to analyze the current signal used for fault diagnosis is conducted through a verified algorithm

during model operation, the amount of additional computation required is small. To evaluate the performance of the LRP algorithm, the performance was evaluated using the MNIST data set. The model used is LE-NET. Figure 4 shows the LRP output for both MNIST data.

4 Experiments and Results

It is possible to analyze the limitations of the current data by checking whether model learning using the current data is possible. In addition, performance analysis with the vibration signal is required. When diagnosing a fault using current data, it is preferable to use a current signal if the same effect as using a vibration signal can be obtained. Current signals and vibration signals were collected in previous studies [11]. The experimental environment to measure vibration signals and current signals is shown in Fig. 5.

The induction motor is used for experiments. In the factory, the commonly used motor is induction motor. So, induction motor is mainly diagnosed. The motor capacity is 200 [W] and it has 4 poles. The LABVIEW is used for implement the measurement.

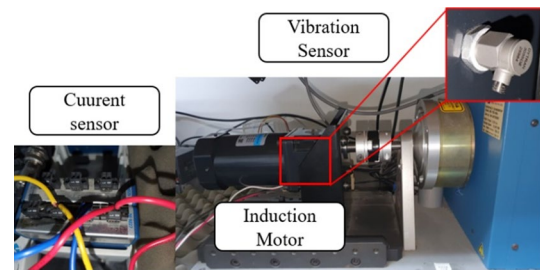


Fig. 5 Vibration and current signal measurement environment

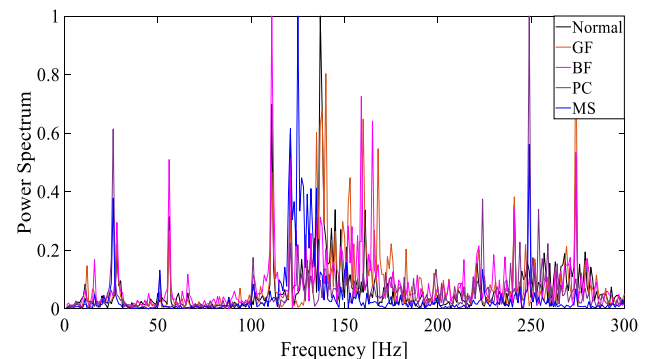


Fig. 6 Vibration FFT when initial fault in each state

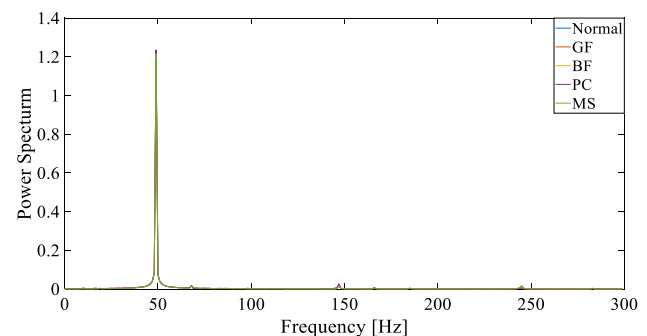


Fig. 7 Current FFT when initial fault in each state

To minimize the difference between the data used for model training, the two data are measured simultaneously. Figures 6 and 7 show the signals of the initial failure state.

Vibration signal has a clear difference according to each state. In this case, motor states are Normal, Gear Fault (GF), Bearing Fault (BF), Pool Connect (PC), Misalignment (MS). On the other hand, there is little difference in the current signal. LE-NET was used to minimize the relevance passed through each layer. The more complex the structure, the more difficult it is to analyze the effect on the input. The output of LE-NET is normal and faulty. Table 1 shows the initial failure state classification performance. Model training

Table 1 5 Class initial fault classifier accuracy

Vibration signal	99.16%
Current signal	20%

Table 2 Pearson coefficients in each state (Vibration)

	Normal	B.F	G.F	M.S	P.C
Normal	1	0.4878	0.5970	0.4815	0.8031
B.F	0.4878	1	0.2498	0.1037	0.2844
G.F	0.5970	0.2498	1	0.4132	0.3485
M.S	0.4815	0.1037	0.4132	1	0.2012
P.C	0.8031	0.2844	0.3485	0.2012	1

Table 3 Pearson coefficients in each state (Current)

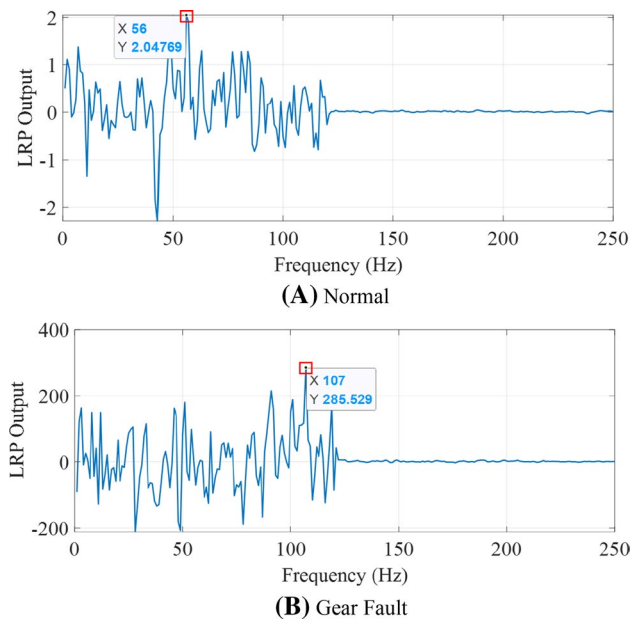
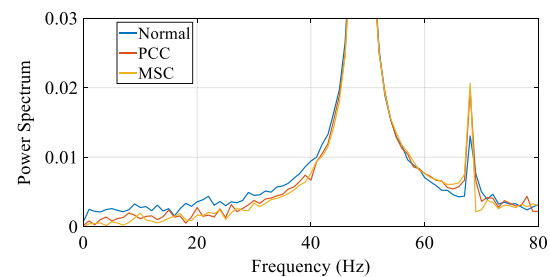
	Normal	B.F	G.F	M.S	P.C
Normal	1	0.9997	0.9999	0.9999	0.9996
B.F	0.9997	1	0.9998	0.9998	0.9997
G.F	0.9999	0.9998	1	0.9997	0.9998
M.S	0.9999	0.9998	0.9997	1	0.9997
P.C	0.9996	0.9999	0.9998	0.9997	1

proceeds through hyperparameter optimization using the genetic algorithm previously studied [12].

In this experiment, the accuracy means the accuracy on the confusion matrix. The deep learning algorithm can classify vibration signals with 99% accuracy even in the initial failure state. However, the initial current data has 20% accuracy. In other words, the model has not been fully trained. The downside of XAI is that it cannot be used unless the model is trained. Because the model has not been trained, it is difficult to trust the validity of the data. A classical correlation coefficient is used to solve this problem. It can analyze patterns and similarities in data. The Pearson correlation coefficient is used to solve this problem. Pearson correlation coefficients in the initial fault data are shown in Tables 2 and 3.

The Pearson correlation coefficient of vibration data is relatively small. However, in the current data, the overall correlation is very close to 1. It can be seen from the model output and the Pearson correlation coefficient that it is desirable to use the current data for initial failure diagnosis. However, learning using current data is possible. Using the current data to train a model to classify normal state and initial gear failure states, the accuracy is 96.67%. Figure 8 shows the LRP output.

The maximum LRP value for this model is 57 HZ in normal condition and 107 HZ in gear failure condition. There is no data difference at 57.107 Hz. This phenomenon can be interpreted as overfitting. LRP analysis can be used to

**Fig. 8** LRP output of binary classifier**Fig. 9** Critical fault current signal**Table 4** Pearson correlation coefficients of critical fault

	Normal	GFC	PCC	MSC
Normal	1	− 0.0046	0.9998	0.9999
GFC	− 0.0046	1	− 0.0064	− 0.0074
PCC	0.9998	− 0.0064	1	0.9997
MSC	0.9999	− 0.0074	0.9997	1

account for the limitations of current data. Additionally, LRP analysis was performed for critical failure conditions. Under the critical error condition, the difference in the current signal becomes apparent. Figure 9 shows the fault current in the critical fault condition.

Table 4 shows the Pearson correlation coefficients in the critical failure state.

As a result of learning the normal, Gear Fault-Current (GFC), Misalignment-Current (MSC), and Pool Connect-Current (PCC) pentagonal classifier including these critical

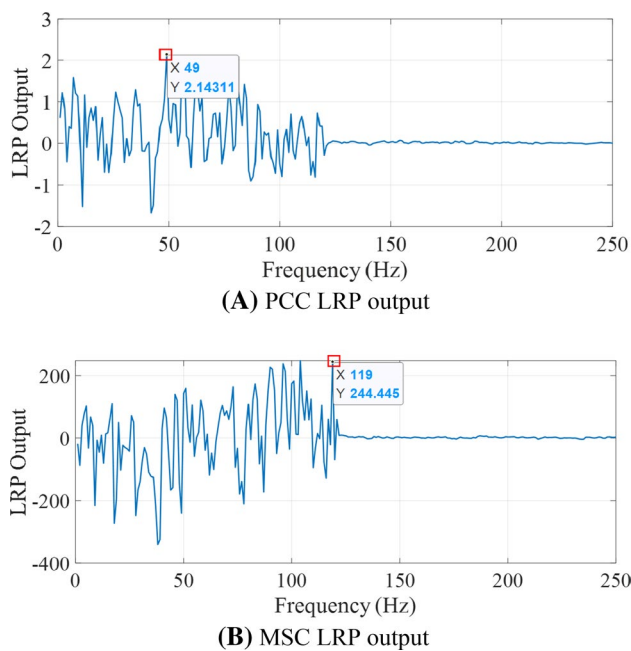


Fig. 10 LRP output of critical fault states

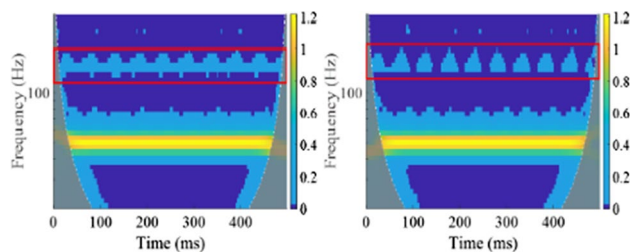


Fig. 11 CWT data

fault states, the accuracy is 97.48%. This means that in the critical failure state, adequate learning is possible because there is a sufficient difference between the data. Existing methods were tested using these failures with a large difference in motor signal. The LRP output of the learning model using these signals is shown in Fig. 10.

It was confirmed that MSC and PCC, which have few characteristics, show a similar trend as in Fig. 8, even in the case of critical failure. However, in the case of a gear failure, it has a clear failure signal, so in the case of a gear failure, it is confirmed that a sufficiently critical failure state can be diagnosed through LRP analysis.

As such, the deep failure state is sufficient data to be used for diagnosis. However, accurate fault diagnosis should be able to diagnose the initial fault condition. A system that is expensive compared to the current signal is required for measuring the vibration signal. This problem can be solved through CWT preprocessing [13]. Figure 11 shows the CWT

normal state current signal and the initial gear failure state data.

Through this study, it was confirmed that the FFT waveform of the current signal is not sufficient for diagnosing the initial fault condition, and that the FFT waveform can be used in the case of a deep fault condition. In addition, it was confirmed that a separate pre-processing process was required to solve this problem.

5 Conclusions and Discussions

A study on diagnosing mechanical faults using current data, which is difficult to express mechanical fault components, was conducted. The current signal is affected by load fluctuations. Current data was found to be difficult to use as initial failure data. Under certain conditions, current data can be learned, but XAI analysis confirmed that this is an overfitting phenomenon. Through XAI analysis, the model found feature points at different points from the classical method. The classical method is difficult to consider all of the numerous motor fault conditions. As such, current data analysis technology based on XAI provides credibility in deep learning. In summary, the analyzed current signal is undesirable to use in the initial fault condition of the motor. Through this results, a study on the effectiveness of current data in the critical fault condition will be studied.

Acknowledgements This work was supported by Korea Electric Power Corporation [Grant number : R21X001-14].

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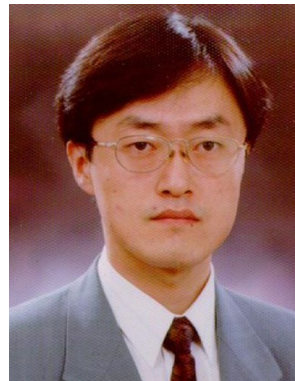
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